



Qualification Pack



Basic Rigger

QP Code: SSD/VSQ/Q0301

Version: 1.0

NSQF Level: 3

Safety Skill Development Foundation || Safety Skill Development Foundation D-507, Light House,
Town Square 1 Sector 82A
Gurugram || email:saurabh@ssdfindia.org



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SSD/VSQ/Q0301: Basic Rigger

Brief Job Description

Basic Rigger is responsible for carrying rigging work, aligning, and anchoring the machinery, using correct rigging technique, attaching loads, controlling the movement of heavy equipment, and ensuring that the rigging is safe for use.

Personal Attributes

He should be physically & mentally fit and should be able to ensure compliances required as per safety and health requirements. He should also be able to provide correct & timely advice on the safety and health issues of construction workers to the management.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSD/VSQ/N0301: Introduction to Basic Rigging](#)
2. [SSD/VSQ/N0302: Safety Standards and Regulations](#)
3. [SSD/VSQ/N0303: Rigging Inspections](#)
4. [SSD/VSQ/N0304: Rigging Equipment and Slings](#)
5. [SSD/VSQ/N0305: Lifting Devices and Below-the-Hook Lifters](#)
6. [SSD/VSQ/N0306: Manual Hoists & Load Handling](#)
7. [SSD/VSQ/N0307: Rigging Techniques and Load Stability](#)
8. [SSD/VSQ/N0308: Contingency Strategies for Emergency Situations](#)
9. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Cross Sectoral
Sub-Sector	Hydrocarbon, Iron & steel, Mining, Power, Automotive, Construction, Chemicals & Petrochemicals, and others.
Occupation	Lifting & Rigging Engineering & Management



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Country	India
NSQF Level	3
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7215.0100
Minimum Educational Qualification & Experience	10th grade pass (or equivalent) with NA of experience OR 9th grade pass with 1-2 Years of experience (1.5 Years) OR 8th grade pass with 3 Years of experience OR Previous relevant Qualification of NSQF Level (2.5) with 1-2 Years of experience (1.5 Years)
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	Nil
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	22/10/2027
NSQC Approval Date	22/10/2024
Version	1.0
Reference code on NQR	QC-03-CO-03364-2024-V1-SSDF
NQR Version	1

Remarks:

NA



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SSD/VSQ/N0301: Introduction to Basic Rigging

Description

The NOS covers the knowledge & skills required by the professional to familiarize with terms & definitions, concepts used in rigging, material & hardware, slings, working load limit and type of slings and their advantages.

Scope

The scope covers the following :

- Familiarization with the terms & definitions used in rigging applications and inspections and principles behind basic rigging theory.
- Different pieces of rigging hardware & applications for various pieces of rigging equipment, sling types and proper use conditions.
- Advantages of different pieces of rigging hardware in unique use cases.
- Working load limit (WLL) load weight calculation, center of gravity for diverse types of slings.
- Safety standards and regulations governing rigging activities.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define key terminologies and basic concepts in rigging. (e.g., hitches, slings, loads).
- PC2.** Explain the fundamental principles of rigging, including weight distribution, load angles, and center of gravity.
- PC3.** Describe the roles and responsibilities of a Basic Rigger.
- PC4.** Identify different types of rigging tools and their applications.
- PC5.** Calculate the weight and center of gravity of a load using basic formulas and estimations.
- PC6.** Identify different types of lifting equipment and their appropriate use for various load types.
- PC7.** Explain the impact of load angles on lifting and rigging safety.
- PC8.** Identify the different types of slings (wire rope, chain, synthetic) and their properties.
- PC9.** Select appropriate slings and rigging hardware for specified lifting tasks.
- PC10.** Demonstrate the correct use of basic hitch configurations for securing loads.
- PC11.** Outline the safety standards and regulations applicable to basic rigging operations.
- PC12.** Identify potential hazards in rigging operations and the appropriate measures to mitigate them.
- PC13.** Apply safety protocols during rigging tasks, including the use of personal protective equipment (PPE).

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Rigging terminologies and definitions, including types of slings, hitches, and hardware.



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- KU2.** Understanding of the principles of weight distribution, center of gravity, and the impact of load angles.
- KU3.** Awareness of different types of rigging equipment, their properties, and limitations.
- KU4.** Basic inspection procedures for rigging equipment.
- KU5.** Safety standards, regulations, and best practices in the rigging industry.
- KU6.** Familiarity with potential hazards in rigging operations and methods to control or eliminate risks.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculating tools.
- GS4.** Understanding & adherence to instructions.
- GS5.** Understanding of safety and adherence to safety related protocols.
- GS6.** Conduct of Professional behavior with co-workers and employees.
- GS7.** Use of personal protective equipment.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Knowledge of organizational hierarchy for availability of tools, equipment and material & its security.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Define key terminologies and basic concepts in rigging. (e.g., hitches, slings, loads).	3	5	-	-
PC2. Explain the fundamental principles of rigging, including weight distribution, load angles, and center of gravity.	3	5	-	-
PC3. Describe the roles and responsibilities of a Basic Rigger.	3	5	-	-
PC4. Identify different types of rigging tools and their applications.	3	5	-	-
PC5. Calculate the weight and center of gravity of a load using basic formulas and estimations.	3	5	-	-
PC6. Identify different types of lifting equipment and their appropriate use for various load types.	3	5	-	-
PC7. Explain the impact of load angles on lifting and rigging safety.	3	5	-	-
PC8. Identify the different types of slings (wire rope, chain, synthetic) and their properties.	3	5	-	-
PC9. Select appropriate slings and rigging hardware for specified lifting tasks.	3	5	-	-
PC10. Demonstrate the correct use of basic hitch configurations for securing loads.	3	5	-	-
PC11. Outline the safety standards and regulations applicable to basic rigging operations.	3	4	-	-
PC12. Identify potential hazards in rigging operations and the appropriate measures to mitigate them.	3	3	-	-
PC13. Apply safety protocols during rigging tasks, including the use of personal protective equipment (PPE).	4	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0301
NOS Name	Introduction to Basic Rigging
Sector	Cross Sectoral
Sub-Sector	
Occupation	Lifting & Rigging Engineering & Management
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



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SSD/VSQ/N0302: Safety Standards and Regulations

Description

The NOS describes the knowledge & skills required by the professional to understand the safety requirements and regulations that govern rigging and lifting operations. It aims to ensure that the Basic Riggers are equipped with the knowledge and skills required to perform rigging tasks in compliance with legal and safety standards. The unit covers various safety regulations, hazard identification, risk management, and compliance procedures specific to rigging and lifting activities.

Scope

The scope covers the following :

- Rigging activities performed by basic Riggers, focusing on adherence to safety standards and regulatory guidelines.
- Understanding industry-specific safety protocols, implementing hazard prevention strategies, and maintaining a safe work environment.
- Use of personal protective equipment (PPE) and ensuring compliance with occupational health and safety (OHS) regulations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and describe the key safety standards and regulations applicable to rigging operations, including OSHA, ANSI, and other relevant standards.
- PC2.** Explain the legal requirements for workplace safety and the responsibilities of a Basic Rigger in maintaining safety compliance.
- PC3.** Demonstrate knowledge of the company's safety policies and procedures related to rigging activities.
- PC4.** Identify potential hazards in rigging and lifting operations, such as load instability, equipment failure, and environmental factors.
- PC5.** Evaluate risks associated with identified hazards and implement strategies to control or mitigate them.
- PC6.** Develop and implement a risk management plan that addresses common hazards in rigging environments.
- PC7.** Ensure proper use of personal protective equipment (PPE) and inspect PPE for compliance with safety standards before use.
- PC8.** Adhere to safety protocols and procedures when performing rigging tasks, including load handling, lifting, and securing activities.
- PC9.** Report safety issues or non-compliance with safety regulations to supervisors or safety officers.
- PC10.** Maintain accurate records of safety inspections, equipment inspections, and incident reports as per regulatory requirements.
- PC11.** Complete safety checklists, hazard assessments, and compliance reports in accordance with industry standards.



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PC12. Communicate safety concerns and compliance issues effectively with team members and supervisors.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Comprehensive understanding of safety standards and regulations, including OSHA, ANSI, and other applicable guidelines.
- KU2.** Company-specific safety policies, safety equipment requirements, and operational procedures.
- KU3.** Familiarity with hazard identification techniques and risk assessment methods for rigging and lifting operations.
- KU4.** Understanding of the proper use, care, and maintenance of personal protective equipment (PPE) and other safety devices.
- KU5.** Awareness of documentation practices, including completing inspection records, safety checklists, and incident reports.
- KU6.** Reporting procedures for non-compliance and safety incidents.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculating tools.
- GS4.** Understanding & adherence to instructions.
- GS5.** Understanding of safety and adherence to safety related protocols.
- GS6.** Conduct of Professional behavior with co-workers and employees.
- GS7.** Use of personal protective equipment.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Strong focus on safety standards, ensuring a safe working environment, and implementing preventative measures.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Identify and describe the key safety standards and regulations applicable to rigging operations, including OSHA, ANSI, and other relevant standards.	3	5	-	-
PC2. Explain the legal requirements for workplace safety and the responsibilities of a Basic Rigger in maintaining safety compliance.	3	5	-	-
PC3. Demonstrate knowledge of the company's safety policies and procedures related to rigging activities.	3	5	-	-
PC4. Identify potential hazards in rigging and lifting operations, such as load instability, equipment failure, and environmental factors.	3	5	-	-
PC5. Evaluate risks associated with identified hazards and implement strategies to control or mitigate them.	3	5	-	-
PC6. Develop and implement a risk management plan that addresses common hazards in rigging environments.	3	5	-	-
PC7. Ensure proper use of personal protective equipment (PPE) and inspect PPE for compliance with safety standards before use.	3	5	-	-
PC8. Adhere to safety protocols and procedures when performing rigging tasks, including load handling, lifting, and securing activities.	3	5	-	-
PC9. Report safety issues or non-compliance with safety regulations to supervisors or safety officers.	3	5	-	-
PC10. Maintain accurate records of safety inspections, equipment inspections, and incident reports as per regulatory requirements.	3	5	-	-
PC11. Complete safety checklists, hazard assessments, and compliance reports in accordance with industry standards.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Communicate safety concerns and compliance issues effectively with team members and supervisors.	5	5	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0302
NOS Name	Safety Standards and Regulations
Sector	Cross Sectoral
Sub-Sector	
Occupation	Lifting & Rigging Engineering & Management
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



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SSD/VSQ/N0303: Rigging Inspections

Description

The NOS describes the knowledge & skills required by the professional to understand the essential competencies required to conduct thorough inspections of rigging equipment, components, and configurations to ensure safe and compliant lifting operations. Identifying, evaluation, and documentation of the condition of various rigging tools and materials to prevent accidents and maintain operational integrity. This unit encompasses pre-use, periodic, and post-use inspections, as well as the application of safety standards for different types of rigging equipment.

Scope

The scope covers the following :

- Inspection of rigging equipments used in lifting and material handling operations, including slings, chains, wire ropes, hooks, and other hardware.
- Identification of defects, assessment of equipment wear and tear, understanding inspection frequency requirements, and documenting inspection results.
- Compliance with industry standards, such as OSHA and ANSI.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify potential hazards in rigging and lifting operations, such as load instability, equipment failure, and environmental factors.
- PC2.** Identify the operational limits and specifications of different rigging tools & devices and develop risk management plan that addresses common hazards in rigging environments.
- PC3.** Identify and describe the key safety standards and regulations applicable to rigging operations, including OSHA, ANSI, and other relevant standards.
- PC4.** Conduct comprehensive risks & hazards identification for rigging tasks and take safety measures.
- PC5.** Perform precise load calculations considering factors such as weight, center of gravity, and balance.
- PC6.** Identify and document any signs of wear, damage, or defects in rigging gear.
- PC7.** Coordinate with team members to ensure synchronized and safe rigging operations.
- PC8.** Develop and practice emergency response plans specific to rigging operations.
- PC9.** Adhere to safety protocols and procedures when performing rigging tasks, including load handling, lifting, and securing activities.
- PC10.** Ensure proper use of personal protective equipment (PPE) and inspect PPE for compliance with safety standards before use.
- PC11.** Raise all the hooks to upper limit switch. Place all controls in the off position. Visually check for dangerous conditions.
- PC12.** Ensure measures that no personnel stand under suspended loads.

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** Inspection procedures and standards for different types of rigging equipment, including slings, chains, wire ropes, and hardware.
- KU2.** Industry-specific safety regulations and guidelines, such as OSHA and ANSI, for conducting rigging inspections.
- KU3.** Equipment-specific wear and damage criteria, including acceptable limits for chain stretch, sling abrasion, and wire rope fraying.
- KU4.** Labeling requirements, identification markings, and load capacity ratings for various rigging equipment.
- KU5.** Environmental factors that can affect equipment integrity and the ability to assess their impact.
- KU6.** Proper tagging, documentation, and reporting procedures for defective or unsafe equipment.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculations.
- GS4.** Understanding & adherence to instructions and safety & compliance standards.
- GS5.** Understanding of safety and adherence to safety related protocols.
- GS6.** Conduct of Professional behavior with co-workers and employees.
- GS7.** Use of personal protective equipment.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Proficiency in maintaining detailed inspection records, completing compliance checklists, and preparing accurate reports.
- GS11.** Focus on safety and compliance standards.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Identify potential hazards in rigging and lifting operations, such as load instability, equipment failure, and environmental factors.	3	5	-	-
PC2. Identify the operational limits and specifications of different rigging tools & devices and develop risk management plan that addresses common hazards in rigging environments.	3	5	-	-
PC3. Identify and describe the key safety standards and regulations applicable to rigging operations, including OSHA, ANSI, and other relevant standards.	3	5	-	-
PC4. Conduct comprehensive risks & hazards identification for rigging tasks and take safety measures.	3	5	-	-
PC5. Perform precise load calculations considering factors such as weight, center of gravity, and balance.	3	5	-	-
PC6. Identify and document any signs of wear, damage, or defects in rigging gear.	3	5	-	-
PC7. Coordinate with team members to ensure synchronized and safe rigging operations.	3	5	-	-
PC8. Develop and practice emergency response plans specific to rigging operations.	3	5	-	-
PC9. Adhere to safety protocols and procedures when performing rigging tasks, including load handling, lifting, and securing activities.	3	5	-	-
PC10. Ensure proper use of personal protective equipment (PPE) and inspect PPE for compliance with safety standards before use.	3	5	-	-
PC11. Raise all the hooks to upper limit switch. Place all controls in the off position. Visually check for dangerous conditions.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Ensure measures that no personnel stand under suspended loads.	5	5	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0303
NOS Name	Rigging Inspections
Sector	Cross Sectoral
Sub-Sector	
Occupation	Lifting & Rigging Engineering & Management
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQF Clearance Date	22/10/2024



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SSD/VSQ/N0304: Rigging Equipment and Slings

Description

The NOS describes the knowledge & skills required by the professional about "Rigging Equipment and Sling", various rigging equipment and sling types used in lifting and material handling operations. This unit equips Basic Riggers with the knowledge and skills required to select, use, and maintain different types of rigging slings and equipment based on load characteristics and safety standards. The NOS covers the properties, applications, and limitations of wire rope slings, chain slings, synthetic slings, and metal mesh slings, as well as rigging hardware such as hooks, shackles, and clips, use of different types of knots, such as bowline, clove hitch, and square knot, with various rope types, including manila, fiber, and synthetic ropes, to ensure proper load securing and handling during rigging operations.

Scope

The scope covers the following :

- Selection and safe use of rigging equipment and sling types in lifting operations.
- Understanding the characteristics of different sling materials, assessing the appropriate usage based on load and environmental conditions, basic checks to ensure the integrity of rigging equipment.
- Use of knots and rigging hardware and accessories to enhance load stability and control during lifting.
- Various knots and their applications for securing loads with different rope types, including manila, fiber, and synthetic ropes.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the different types of rigging slings, including wire rope slings, chain slings, synthetic web slings, round slings, and metal mesh slings.
- PC2.** Describe the properties, strengths, and limitations of each sling type in terms of load capacity, flexibility, and durability.
- PC3.** Select the appropriate sling type for different load characteristics, lifting conditions, and environmental factors.
- PC4.** Evaluate the load weight, center of gravity, and lifting requirements to choose the correct rigging equipment and configuration.
- PC5.** Select and attach appropriate rigging hardware, such as hooks, shackles, clips, and turnbuckles, to enhance load stability and control.
- PC6.** Demonstrate the correct use of sling hitches, including vertical, choker, and basket hitches, for effective load handling.
- PC7.** Select and tie the appropriate knot, such as bowline, clove hitch, square knot, or double half hitch, using manila, fiber, or synthetic ropes for securing loads during rigging operations.
- PC8.** Understand the properties and limitations of different types of ropes, such as manila, fiber, and synthetic, and how these properties impact knot selection and load securing.
- PC9.** Inspect rigging slings, equipment, and ropes before use to identify any signs of wear, damage, or deformation.



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- PC10.** Perform basic maintenance on rigging slings, equipment, and ropes as per manufacturer guidelines and industry standards.
- PC11.** Remove from service and tag out any rigging slings, equipment, or ropes found to be unsafe or defective during inspection.
- PC12.** Use rigging hardware such as hooks, shackles, eye bolts, and clips according to manufacturer specifications and safety guidelines.
- PC13.** Calculate load angles and stress factors when using multiple rigging slings, ropes, or attachments to ensure balanced lifting.
- PC14.** Employ proper rigging techniques, including tying appropriate knots and securing with rigging hardware, to prevent sling slippage, rope untying, load swinging, or tipping during lifting operations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different rigging sling types, their properties, and appropriate applications based on load characteristics and lifting conditions.
- KU2.** Rigging hardware, such as hooks, shackles, wire rope clips, and turnbuckles, and their proper usage in lifting operations.
- KU3.** Safety standards and regulations related to rigging equipment selection and usage, including load ratings and identification markings.
- KU4.** Sling and rope inspection procedures and maintenance practices to ensure equipment safety and longevity.
- KU5.** Sling hitch configurations (vertical, choker, basket) and their impact on load stability and distribution.
- KU6.** Different types of knots, including bowline, clove hitch, square knot, and double half hitch, and their use with manila, fiber, and synthetic ropes.
- KU7.** Properties of various rope materials (e.g., manila, fiber, synthetic) and their impact on load handling, knot stability, and securing effectiveness.
- KU8.** Impact of environmental factors, such as temperature, moisture, and chemical exposure, on rigging sling and rope performance and safety.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculations.
- GS4.** Understanding & adherence to instructions.
- GS5.** Understanding of safety and adherence to safety related protocols.
- GS6.** Conduct of Professional behavior with co-workers and employees.
- GS7.** Use of personal protective equipment.



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- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Understand lifting challenges and appropriate rigging and knot-tying solutions.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Identify the different types of rigging slings, including wire rope slings, chain slings, synthetic web slings, round slings, and metal mesh slings.	3	4	-	-
PC2. Describe the properties, strengths, and limitations of each sling type in terms of load capacity, flexibility, and durability.	3	4	-	-
PC3. Select the appropriate sling type for different load characteristics, lifting conditions, and environmental factors.	3	4	-	-
PC4. Evaluate the load weight, center of gravity, and lifting requirements to choose the correct rigging equipment and configuration.	3	4	-	-
PC5. Select and attach appropriate rigging hardware, such as hooks, shackles, clips, and turnbuckles, to enhance load stability and control.	3	4	-	-
PC6. Demonstrate the correct use of sling hitches, including vertical, choker, and basket hitches, for effective load handling.	3	4	-	-
PC7. Select and tie the appropriate knot, such as bowline, clove hitch, square knot, or double half hitch, using manila, fiber, or synthetic ropes for securing loads during rigging operations.	3	4	-	-
PC8. Understand the properties and limitations of different types of ropes, such as manila, fiber, and synthetic, and how these properties impact knot selection and load securing.	3	4	-	-
PC9. Inspect rigging slings, equipment, and ropes before use to identify any signs of wear, damage, or deformation.	3	4	-	-
PC10. Perform basic maintenance on rigging slings, equipment, and ropes as per manufacturer guidelines and industry standards.	3	4	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Remove from service and tag out any rigging slings, equipment, or ropes found to be unsafe or defective during inspection.	2	5	-	-
PC12. Use rigging hardware such as hooks, shackles, eye bolts, and clips according to manufacturer specifications and safety guidelines.	2	5	-	-
PC13. Calculate load angles and stress factors when using multiple rigging slings, ropes, or attachments to ensure balanced lifting.	3	5	-	-
PC14. Employ proper rigging techniques, including tying appropriate knots and securing with rigging hardware, to prevent sling slippage, rope untying, load swinging, or tipping during lifting operations.	3	5	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0304
NOS Name	Rigging Equipment and Slings
Sector	Cross Sectoral
Sub-Sector	
Occupation	Lifting & Rigging Engineering & Management
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



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SSD/VSQ/N0305: Lifting Devices and Below-the-Hook Lifters

Description

The NOS describes the knowledge & skills required by the professional to safely operate, select, and maintain various lifting devices and below-the-hook lifting equipment. This NOS covers the principles, safe usage, and limitations of lifting beams, spreader beams, clamps, lifting magnets, vacuum lifters, and other below-the-hook devices. Basic Riggers will learn how to choose the appropriate lifting device based on load characteristics and lifting conditions, ensuring safe and efficient handling of loads.

Scope

The scope covers the following :

- Selection, inspection, and safe operation of lifting devices and below-the-hook lifters used in material handling and lifting operations.
- Understanding the applications, strengths, and limitations of each type of lifting device, as well as conducting inspections to ensure compliance with safety standards.
- Integration of these devices with other rigging components to ensure stable and safe lifting operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify different types of lifting devices and below-the-hook lifters, including lifting beams, spreader beams, plate clamps, lifting magnets, and vacuum lifters.
- PC2.** Describe the properties, applications, and limitations of each lifting device in relation to load characteristics and lifting conditions.
- PC3.** Select the appropriate lifting device based on load requirements, lifting points, and environmental factors.
- PC4.** Demonstrate the proper use of lifting beams and spreader beams, ensuring balanced load distribution and stability.
- PC5.** Operate clamps, lifting magnets, and vacuum lifters in accordance with manufacturer specifications and safety guidelines.
- PC6.** Attach and secure lifting devices to loads using appropriate rigging techniques to prevent load slippage or tipping.
- PC7.** Conduct pre-use inspections of lifting devices and below-the-hook lifters to identify defects, wear, or other signs of damage.
- PC8.** Perform basic maintenance on lifting devices as per manufacturer guidelines to ensure safe and continued functionality.
- PC9.** Remove from service and tag any lifting device found to be defective or unsafe and report the issue to the supervisor for further action.
- PC10.** Calculate load weights, center of gravity, and load distribution when using below-the-hook lifters to ensure stability.
- PC11.** Assess the impact of load angle factors, load integrity, and environmental conditions on the safe operation of lifting devices.



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- PC12.** Apply appropriate rigging configurations and hitch types to maintain control and stability during lifting operations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different types of lifting devices and below-the-hook lifters, including lifting beams, spreader beams, clamps, magnets, and vacuum lifters.
- KU2.** Principles and mechanics of load handling, including load weight estimation, center of gravity calculation, and load distribution.
- KU3.** Safety standards and regulatory guidelines, such as OSHA and ANSI, governing the use of lifting devices and below-the-hook lifters.
- KU4.** Inspection procedures, maintenance practices, and tagging requirements for defective or unsafe lifting equipment.
- KU5.** Environmental factors that can impact the safe use of lifting devices, including temperature, wind, and humidity.
- KU6.** Load dynamics and the impact of load angle, stress factors, and stability considerations during lifting operations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculations.
- GS4.** Understanding & adherence to instructions.
- GS5.** Focus on adhering to safety standards and guidelines.
- GS6.** Conduct of Professional behavior with co-workers and employees.
- GS7.** Use of personal protective equipment.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Knowledge of organizational hierarchy for availability of tools, equipment & material & its security.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Identify different types of lifting devices and below-the-hook lifters, including lifting beams, spreader beams, plate clamps, lifting magnets, and vacuum lifters.	3	5	-	-
PC2. Describe the properties, applications, and limitations of each lifting device in relation to load characteristics and lifting conditions.	3	5	-	-
PC3. Select the appropriate lifting device based on load requirements, lifting points, and environmental factors.	3	5	-	-
PC4. Demonstrate the proper use of lifting beams and spreader beams, ensuring balanced load distribution and stability.	3	5	-	-
PC5. Operate clamps, lifting magnets, and vacuum lifters in accordance with manufacturer specifications and safety guidelines.	3	5	-	-
PC6. Attach and secure lifting devices to loads using appropriate rigging techniques to prevent load slippage or tipping.	3	5	-	-
PC7. Conduct pre-use inspections of lifting devices and below-the-hook lifters to identify defects, wear, or other signs of damage.	3	5	-	-
PC8. Perform basic maintenance on lifting devices as per manufacturer guidelines to ensure safe and continued functionality.	3	5	-	-
PC9. Remove from service and tag any lifting device found to be defective or unsafe and report the issue to the supervisor for further action.	3	5	-	-
PC10. Calculate load weights, center of gravity, and load distribution when using below-the-hook lifters to ensure stability.	3	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Assess the impact of load angle factors, load integrity, and environmental conditions on the safe operation of lifting devices.	5	5	-	-
PC12. Apply appropriate rigging configurations and hitch types to maintain control and stability during lifting operations.	5	5	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0305
NOS Name	Lifting Devices and Below-the-Hook Lifters
Sector	Cross Sectoral
Sub-Sector	
Occupation	Lifting & Rigging Engineering & Management
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

SSD/VSQ/N0306: Manual Hoists & Load Handling

Description

The NOS describes the knowledge & skills required by the professional for proper selection, operation, and maintenance of manual hoists and the safe handling of loads during lifting operations. This NOS equips Basic Riggers with the knowledge and skills to use manual lifting devices such as chain hoists, lever hoists, and hand-operated winches. It covers the principles of load handling, safe lifting practices, and compliance with industry standards. The Basic Riggers will be able to demonstrate competence in selecting the appropriate manual hoist, performing pre-use inspections, and handling loads safely.

Scope

The scope covers the following :

- Use of manual hoists in lifting and material handling operations.
- Understanding the characteristics, capacities, and limitations of manual hoists, performing pre-use inspections and maintenance, and applying safe load handling techniques.
- Load estimation, hoist configurations, and hazard management to ensure safe and efficient lifting operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify different types of manual hoists, including chain hoists, lever hoists, and hand-operated winches, and describe their properties and applications.
- PC2.** Determine the appropriate manual hoist type based on load weight, lifting height, and environmental conditions.
- PC3.** Explain the limitations of manual hoists in terms of load capacity, operational distance, and required force.
- PC4.** Operate manual hoists safely, ensuring that loads are lifted and lowered in a controlled and stable manner.
- PC5.** Use appropriate lifting techniques, including correct positioning and alignment of the hoist, to prevent load sway and imbalance.
- PC6.** Attach loads securely to the hoist using appropriate rigging techniques and hardware to ensure safe lifting.
- PC7.** Perform pre-use inspections of manual hoists to check for signs of wear, damage, or malfunction, such as chain defects, hook deformation, or gear damage.
- PC8.** Inspect hooks, chains, and other components to ensure they meet safety standards and manufacturer guidelines.
- PC9.** Conduct routine maintenance of manual hoists, including lubrication and adjustment of mechanical components, as recommended by the manufacturer.
- PC10.** Estimate load weight and center of gravity to ensure safe and balanced lifting with manual hoists.
- PC11.** Identify potential hazards, such as overhead obstacles or unstable ground, and implement measures to mitigate these risks during load handling.



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- PC12.** Apply appropriate lifting configurations, including vertical, angled, or multiple hoist setups, to ensure load stability and control.
- PC13.** Document inspection results, maintenance activities, and any identified issues according to company procedures and industry standards.
- PC14.** Report safety concerns, equipment defects, or incidents to supervisors and safety officers for prompt resolution.
- PC15.** Ensure compliance with OSHA, ANSI, and other regulatory guidelines related to the use of manual hoists and load handling.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different types of manual hoists, their applications, and limitations in lifting operations.
- KU2.** Manual hoist mechanics, including load chains, hooks, gears, and braking systems.
- KU3.** Industry safety standards and regulatory guidelines governing the use of manual hoists, including load capacity and safety factor requirements.
- KU4.** Inspection procedures, including identifying defects, wear, and malfunctioning components in manual hoists.
- KU5.** Proper rigging techniques for securely attaching loads to manual hoists.
- KU6.** Risk management strategies to handle potential hazards during manual hoist operations and load handling.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculations
- GS4.** Understanding & adherence to instructions.
- GS5.** Understanding of safety and adherence to safety related protocols with industry safety standards and regulatory guidelines.
- GS6.** Conduct of Professional behavior with co-workers and employees.
- GS7.** Use of personal protective equipment.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Knowledge of organizational hierarchy for availability of tools, equipment and material & its security.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Identify different types of manual hoists, including chain hoists, lever hoists, and hand-operated winches, and describe their properties and applications.	3	4	-	-
PC2. Determine the appropriate manual hoist type based on load weight, lifting height, and environmental conditions.	3	4	-	-
PC3. Explain the limitations of manual hoists in terms of load capacity, operational distance, and required force.	3	4	-	-
PC4. Operate manual hoists safely, ensuring that loads are lifted and lowered in a controlled and stable manner.	3	4	-	-
PC5. Use appropriate lifting techniques, including correct positioning and alignment of the hoist, to prevent load sway and imbalance.	3	4	-	-
PC6. Attach loads securely to the hoist using appropriate rigging techniques and hardware to ensure safe lifting.	3	4	-	-
PC7. Perform pre-use inspections of manual hoists to check for signs of wear, damage, or malfunction, such as chain defects, hook deformation, or gear damage.	3	4	-	-
PC8. Inspect hooks, chains, and other components to ensure they meet safety standards and manufacturer guidelines.	3	4	-	-
PC9. Conduct routine maintenance of manual hoists, including lubrication and adjustment of mechanical components, as recommended by the manufacturer.	3	4	-	-
PC10. Estimate load weight and center of gravity to ensure safe and balanced lifting with manual hoists.	3	4	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Identify potential hazards, such as overhead obstacles or unstable ground, and implement measures to mitigate these risks during load handling.	2	4	-	-
PC12. Apply appropriate lifting configurations, including vertical, angled, or multiple hoist setups, to ensure load stability and control.	2	4	-	-
PC13. Document inspection results, maintenance activities, and any identified issues according to company procedures and industry standards.	2	4	-	-
PC14. Report safety concerns, equipment defects, or incidents to supervisors and safety officers for prompt resolution.	2	4	-	-
PC15. Ensure compliance with OSHA, ANSI, and other regulatory guidelines related to the use of manual hoists and load handling.	2	4	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0306
NOS Name	Manual Hoists & Load Handling
Sector	Cross Sectoral
Sub-Sector	
Occupation	Lifting & Rigging Engineering & Management
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

SSD/VSQ/N0307: Rigging Techniques and Load Stability

Description

The NOS describes the knowledge & skills required by the professional to ensure safe and stable handling of loads using various rigging techniques. This NOS describes load characteristics, apply appropriate rigging configurations, and ensure load stability throughout lifting operations with understanding of load dynamics, managing sling angles, and utilizing hitch types to control load movement and prevent accidents during rigging and lifting activities.

Scope

The scope covers the following :

- Application of rigging techniques and configurations used in lifting and material handling operations to ensure load stability.
- Selection of appropriate rigging methods based on load characteristics, employing different types of hitches, and managing load angles to maintain balance and control.
- Identification of potential hazards related to load stability and implementing risk management strategies to prevent tipping, slipping, or other stability-related issues during lifting operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define key rigging concepts, including load angles, center of gravity, load weight distribution, and the effect of sling angles on load stability.
- PC2.** Explain the principles of load dynamics, including how forces and movements affect load stability during lifting operations.
- PC3.** Identify different rigging techniques such as vertical, choker, and basket hitches, and describe their impact on load handling and stability.
- PC4.** Select and apply the appropriate hitch configurations (vertical, choker, basket) based on load characteristics and lifting requirements.
- PC5.** Calculate and manage sling angles to minimize stress and ensure balanced load lifting.
- PC6.** Use multiple slings, spreader beams, or lifting beams to distribute the load evenly and prevent tipping or swinging.
- PC7.** Assess load stability by determining the center of gravity and applying rigging techniques to keep the load level during lifting.
- PC8.** Monitor load movement and make necessary adjustments to sling tension and configurations to maintain load control.
- PC9.** Implement safety measures, such as tag lines or secondary rigging, to control load movement and prevent unwanted load shifts.
- PC10.** Identify potential hazards associated with load stability, such as imbalanced loads, swinging, and load tipping, and assess their impact on lifting operations.
- PC11.** Apply risk management strategies to mitigate identified hazards, including adjusting rigging configurations and using additional rigging equipment.



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- PC12.** Communicate stability-related hazards and control measures to team members and supervisors to ensure safe lifting operations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different rigging techniques and their applications in maintaining load stability.
- KU2.** Principles of load dynamics, including the impact of load weight, center of gravity, and sling angles on load stability.
- KU3.** Hitch configurations (vertical, choker, basket) and their effect on load movement and control.
- KU4.** Calculating sling angles, load forces, and stress factors to optimize rigging configurations.
- KU5.** Potential hazards related to load stability and effective risk management strategies to prevent accidents during lifting operations.
- KU6.** Proper use of additional rigging equipment, such as spreader beams or lifting beams, to ensure even load distribution and stability.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculations.
- GS4.** Understanding & adherence to instructions.
- GS5.** Understanding of safety and adherence to safety related protocols with industry safety standards and regulatory guidelines.
- GS6.** Conduct of Professional behavior with co-workers and employees.
- GS7.** Use of personal protective equipment.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Knowledge of organizational hierarchy for availability of tools, equipment & material & its security.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Define key rigging concepts, including load angles, center of gravity, load weight distribution, and the effect of sling angles on load stability.	3	5	-	-
PC2. Explain the principles of load dynamics, including how forces and movements affect load stability during lifting operations.	3	5	-	-
PC3. Identify different rigging techniques such as vertical, choker, and basket hitches, and describe their impact on load handling and stability.	3	5	-	-
PC4. Select and apply the appropriate hitch configurations (vertical, choker, basket) based on load characteristics and lifting requirements.	3	5	-	-
PC5. Calculate and manage sling angles to minimize stress and ensure balanced load lifting.	3	5	-	-
PC6. Use multiple slings, spreader beams, or lifting beams to distribute the load evenly and prevent tipping or swinging.	3	5	-	-
PC7. Assess load stability by determining the center of gravity and applying rigging techniques to keep the load level during lifting.	3	5	-	-
PC8. Monitor load movement and make necessary adjustments to sling tension and configurations to maintain load control.	3	5	-	-
PC9. Implement safety measures, such as tag lines or secondary rigging, to control load movement and prevent unwanted load shifts.	3	5	-	-
PC10. Identify potential hazards associated with load stability, such as imbalanced loads, swinging, and load tipping, and assess their impact on lifting operations.	3	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Apply risk management strategies to mitigate identified hazards, including adjusting rigging configurations and using additional rigging equipment.	5	5	-	-
PC12. Communicate stability-related hazards and control measures to team members and supervisors to ensure safe lifting operations.	5	5	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0307
NOS Name	Rigging Techniques and Load Stability
Sector	Cross Sectoral
Sub-Sector	
Occupation	Lifting & Rigging Engineering & Management
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

SSD/VSQ/N0308: Contingency Strategies for Emergency Situations

Description

The NOS describes the knowledge & skills required by the professional to respond effectively to emergency situations that may arise during rigging and lifting operations. This NOS focuses on developing contingency plans, identifying potential emergency scenarios, and implementing appropriate response actions to protect personnel, equipment, and the work environment. It ensures that riggers are prepared to handle unexpected events such as equipment failure, load instability, or environmental hazards.

Scope

The scope covers the following :

- Identification of potential emergency scenarios in rigging operations, development of contingency strategies, and execution of emergency response plans.
- Understanding emergency protocols, effective communication during emergencies, and coordination with safety personnel and first responders.
- Conducting emergency drills and maintaining awareness of site-specific emergency procedures to ensure preparedness and safety.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify potential emergency scenarios that could arise during rigging operations, such as equipment failure, load instability, structural collapse, or environmental hazards.
- PC2.** Assess the impact of identified emergency scenarios on personnel safety, equipment integrity, and the surrounding environment.
- PC3.** Evaluate the probability and severity of each emergency scenario to prioritize response strategies.
- PC4.** Develop contingency plans for handling identified emergency scenarios, including clear procedures for equipment shutdown, load securing, and evacuation.
- PC5.** Establish roles and responsibilities for personnel during emergency situations, ensuring that all team members are aware of their duties and actions.
- PC6.** Create communication protocols for emergency situations, including notification of supervisors, safety personnel, and first responders.
- PC7.** Execute emergency response actions in accordance with the developed contingency plan, ensuring the safety of personnel and equipment.
- PC8.** Secure unstable loads or equipment to prevent further hazards and minimize the risk of injury or damage.
- PC9.** Coordinate with safety personnel and first responders to facilitate a quick and effective emergency response.
- PC10.** Conduct regular emergency drills to test the effectiveness of contingency plans and ensure personnel are familiar with emergency procedures.
- PC11.** Provide training on emergency response actions, including equipment shutdown, evacuation, and hazard control.



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- PC12.** Review and update contingency plans based on the outcomes of emergency drills and feedback from participants.
- PC13.** Conduct a post-emergency evaluation to assess the effectiveness of the response actions and identify areas for improvement.
- PC14.** Document the emergency incident, including the sequence of events, actions taken, and any injuries or damages sustained.
- PC15.** Implement corrective actions and update contingency plans based on evaluation findings and lessons learned.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Potential emergency scenarios in rigging and lifting operations, including equipment failure, load instability, and environmental hazards.
- KU2.** Components of effective contingency plans, including emergency response procedures, roles, and communication protocols.
- KU3.** Safety standards and regulatory guidelines related to emergency preparedness and response in rigging operations.
- KU4.** Emergency response techniques, such as equipment shutdown, load securing, and personnel evacuation.
- KU5.** Awareness of coordination procedures with safety personnel, first responders, and other emergency services.
- KU6.** Post-emergency evaluation methods and documentation practices to improve future response strategies.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Reading & writing in the language of communication.
- GS2.** Effective verbal communication in local language.
- GS3.** Basic mathematics and calculations.
- GS4.** Understanding & adherence to instructions.
- GS5.** Safety and adherence to safety related protocols with industry safety standards and regulatory guidelines.
- GS6.** Conduct of Professional behavior with co-workers and employees.
- GS7.** Use of personal protective equipment.
- GS8.** Work site rules & security, fire safety protocols, first aid & procedure under emergency conditions.
- GS9.** Housekeeping and disposal of waste materials.
- GS10.** Organizational hierarchy for availability of tools, equipment & material & its security.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Identify potential emergency scenarios that could arise during rigging operations, such as equipment failure, load instability, structural collapse, or environmental hazards.	3	4	-	-
PC2. Assess the impact of identified emergency scenarios on personnel safety, equipment integrity, and the surrounding environment.	3	4	-	-
PC3. Evaluate the probability and severity of each emergency scenario to prioritize response strategies.	3	4	-	-
PC4. Develop contingency plans for handling identified emergency scenarios, including clear procedures for equipment shutdown, load securing, and evacuation.	3	4	-	-
PC5. Establish roles and responsibilities for personnel during emergency situations, ensuring that all team members are aware of their duties and actions.	3	4	-	-
PC6. Create communication protocols for emergency situations, including notification of supervisors, safety personnel, and first responders.	3	4	-	-
PC7. Execute emergency response actions in accordance with the developed contingency plan, ensuring the safety of personnel and equipment.	3	4	-	-
PC8. Secure unstable loads or equipment to prevent further hazards and minimize the risk of injury or damage.	3	4	-	-
PC9. Coordinate with safety personnel and first responders to facilitate a quick and effective emergency response.	3	4	-	-
PC10. Conduct regular emergency drills to test the effectiveness of contingency plans and ensure personnel are familiar with emergency procedures.	3	4	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Provide training on emergency response actions, including equipment shutdown, evacuation, and hazard control.	2	4	-	-
PC12. Review and update contingency plans based on the outcomes of emergency drills and feedback from participants.	2	4	-	-
PC13. Conduct a post-emergency evaluation to assess the effectiveness of the response actions and identify areas for improvement.	2	4	-	-
PC14. Document the emergency incident, including the sequence of events, actions taken, and any injuries or damages sustained.	2	4	-	-
PC15. Implement corrective actions and update contingency plans based on evaluation findings and lessons learned.	2	4	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSD/VSQ/N0308
NOS Name	Contingency Strategies for Emergency Situations
Sector	Cross Sectoral
Sub-Sector	
Occupation	Lifting & Rigging Engineering & Management
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The assessment criteria given in the Qualification Pack Basic Rigger by the SSDF.
2. Assessments can be carried out with pen-paper as well as online.
3. Assessment will be carried by certified assessor and affiliated Assessment Agency only.
4. Each NOS for its Performance Criteria (PC) has been assigned marks proportional to its importance. Proportion of marks for Theory and Practical has been marked NOS wise.
5. Questions on practical & theory will be formed in such a way as to provide outcome on maximum Performance Criteria and in proportional way within the NOS.
6. The assessment for the theory part will be based on written questions (short question, multiple choice & viva, or a combination of them) created/approved by the SSDF.
7. The assessment for the practical part will be based on practical conducted for trainees. In case of remote/on-line assessments, the practical's can be carried through proctors or practical questions formulated based on pictorially represented logical questions (based on pictures of practical & logical steps) created/approved by the SSDF.



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8. The passing and grading criteria of each NOS & cumulative for QP will be as follows:-

a. 70% or more than 70% - Grade A

b. 60% or more than 60% but less than 70% - Grade B

c. 50% or more than 50% but less than 60% - Grade C

d. Less than 50% - Grade Fail

e. If an individual gets less than 50% and 35% or more in one NOS and; overall, 50% or more; individual will be considered pass with grade "C" irrespective of overall marks.

f. Individuals getting less than 50% in more than one NOS and getting overall marks 50% or more in QP will be put in grade Fail.

9. Assessment will be for complete QP and the trainees can be given additional chance to appear in re-assessment for improvements, based on decision by SSDF.

10. In case a trainee fails in one or two NOSs, re-assessments in the failing NOSs can be allowed in the next 3 months from the date of publication of results. Otherwise, the failed candidate will have to re-appear in all NOSs.

11. Minimum overall duration of assessment will be six hours.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSD/VSQ/N0301.Introduction to Basic Rigging	40	60	0	0	100	9
SSD/VSQ/N0302.Safety Standards and Regulations	40	60	0	0	100	8
SSD/VSQ/N0303.Rigging Inspections	40	60	0	0	100	8



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSD/VSQ/N0304.Rigging Equipment and Slings	40	60	0	0	100	8
SSD/VSQ/N0305.Lifting Devices and Below-the-Hook Lifters	40	60	0	0	100	17
SSD/VSQ/N0306.Manual Hoists & Load Handling	40	60	0	0	100	17
SSD/VSQ/N0307.Rigging Techniques and Load Stability	40	60	0	0	100	17
SSD/VSQ/N0308.Contingency Strategies for Emergency Situations	40	60	0	0	100	8
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	8
Total	340	510	-	-	850	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.